

CHE 453: Fundamentals of Electrochemistry

Course Syllabus – Fall 2024

Class meetings:	Thursday 15:00 - 17:30	EIB 271
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Instructor:

Brian P. Chaplin, Ph.D.: 154 EIB, (312) 996-0288 (office), chaplin@uic.edu
Office hours: TBA; or by appointment

Objectives and Prerequisites

Introduction to the fundamentals of electrochemistry and its application in a variety of technologies (i.e., electrolysis cells, batteries, fuel cells). Includes methods for the analysis of cells using electrochemical techniques and analytical and numerical modeling to interpret results.

Prerequisite: Graduate or advanced undergraduate standing

Required Text and Supplemental Materials

Required Text: Bard, A. J.; Faulkner, L. R., *Electrochemical Methods: Fundamentals and Applications*. Second ed.; John Wiley and Sons: New Jersey, 2001; p 833. ISBN: 978-0-470-45253-0

Supplemental Materials:

- Bockris, J. O. M.; Reddy, A. K.; Gamba-Aldeco, M., *Modern Electrochemistry*. 2nd ed.; John Wiley and Sons: New Jersey, 2004; p 647.
- Newman, J.; Balsara, N.P., *Electrochemical Systems*. 4th ed.; John Wiley and Sons: New Jersey, 2021; p 577.
- A. Lasia, Electrochemical Impedance Spectroscopy and Its Applications, In: *Modern Aspects of Electrochemistry*, B. E. Conway, J. Bockris, and R.E. White, Eds., Kluwer Academic/Plenum Publishers, New York, 1999, Vol. 32, p. 143-248.
- Orazem, M. E.; Tribollet, B., *Electrochemical Impedance Spectroscopy*. John Wiley and Sons: New Jersey, 2008; p 525.

Required Technology

Students will need regular access to a personal computer that runs on a broadband Internet connection. Please contact ACCC (LTS@uic.edu) to request and borrow a laptop or be assigned a hotspot for the semester if you do not have the required hardware and internet capabilities.

If necessary we may use Zoom for live video class meetings. Please make sure you are able to use the software during the scheduled class times.

Grading policy

The course grade will be based on the following:

- Exam 1—35%
- Exam 2—35%
- Assignments, Laboratories, and Quizzes—10%
- Term Project Presentation—20%

Attendance policy

1. Attendance at class meetings is highly recommended.
2. If you miss a class meeting, you are responsible for the course material covered and compliance with any announcements made. Most handouts will be posted online.
3. Attendance at examination periods is mandatory. If you miss an examination, you will receive a zero for that examination. Makeup exams will **ONLY** be given, in the case of an extreme emergency. I will make all decisions on this matter on a case-by-case basis.

Assigned Work

In this course, all assigned work, unless otherwise stated, should be completed and handed in individually. Evidence of direct copying will be addressed in accordance with the University's Academic Integrity policies (see below). Required assignments will be due before a given time on the due date. You are asked to prepare all work in a professional manner. Late work assignments will **NOT** be accepted.

Solutions for homework problems will be available for student review and will be posted on Black Board after the assignment due date. Students are encouraged to **first** compare homework and the correct solution to find the source of errors in their calculations. If there is still some confusion, the student is encouraged to meet with me. Office hours provide an ideal opportunity to discuss such matters.

Term Project and Presentation

Each student is required to give an oral presentation on a topic assigned by the instructor. The presentation should review current research related to a specific electrochemical technology or a scientific question that advances the fundamental understanding of an electrochemical process. Additional details will be provided later in the semester.

Oral presentations will be on **November 21, 2024** and must be 15-20 minutes in length.

Academic Integrity

UIC is an academic community committed to providing an environment in which research, learning, and scholarship can flourish and in which all endeavors are guided by academic and professional integrity. In this community, all members including faculty, administrators, staff, and students alike share the responsibility to uphold the highest standards of academic honesty and quality of academic work so that such a collegial and productive environment exists.

As a student and member of the UIC community, you are expected to adhere to the Community Standards (<https://dos.uic.edu/community-standards/>) of integrity, accountability, and respect in all of your academic endeavors. When accusations of academic dishonesty (<https://dos.uic.edu/community-standards/academic-integrity/>) occur, the Office of the Dean of Students investigates and adjudicates

suspected violations of this student code. Unacceptable behavior includes cheating, unauthorized collaboration, fabrication or falsification, plagiarism, multiple submissions without instructor permission, using unauthorized study aids, coercion regarding grading or evaluation of coursework, and facilitating academic misconduct. Please review the UIC Student Disciplinary Policy (<https://dos.uic.edu/wp-content/uploads/sites/262/2018/10/DOS-Student-Disciplinary-Policy-2018-2019-FINAL.pdf>) for additional information about the process by which instances of academic misconduct are handled towards the goal of developing responsible student behavior.

By submitting your assignments for grading you acknowledge these terms, you declare that your work is solely your own, and you promise that, unless authorized by the instructor or proctor, you have not communicated with anyone in any way during an exam or other online assessment. Let's embrace what it means to be a UIC community member and together be committed to the values of integrity.

Inclusive Learning Environment

UIC values diversity and inclusion. Regardless of age, disability, ethnicity, race, gender, gender identity, sexual orientation, socioeconomic status, geographic background, religion, political ideology, language, or culture, we expect all members of this class to contribute to a respectful, welcoming, and inclusive environment for every other member of our class. If there are aspects of the instruction or design of this course that result in barriers to your inclusion, engagement, accurate assessment or achievement, please notify me as soon as possible.

UIC is committed to full inclusion and participation of people with disabilities in all aspects of university life. If you face or anticipate disability-related barriers while at UIC, please connect with the Disability Resource Center (DRC) at drc.uic.edu, via email at drc@uic.edu, or call (312) 413-2183 to create a plan for reasonable accommodations. In order to receive accommodations, you will need to disclose the disability to the DRC, complete an interactive registration process with the DRC, and provide me with a Letter of Accommodation (LOA). Upon receipt of a LOA, I will gladly work with you and the DRC to implement approved accommodations.

Course Topics (subject to change)

- Introduction and overview of electrochemical cells and electrode processes (Ch. 1 Bard & Notes)
- Thermodynamics of electrochemical cells (Ch. 2 Bard & Notes)
- Reference Electrodes (Notes)
- Kinetics of electrochemical reactions (Ch. 3 Bard & Notes)
- Mass transfer in electrochemical systems (Ch. 4 Bard & Notes)
- Adsorption (Ch. 13 Bard & Notes)
- Electrochemical methods including potential step and sweep methods and hydrodynamic methods. (Ch. 5, 6, 9 Bard & Notes)
- Modeling electrochemical reactions/methods (Notes)
- Electrochemical impedance spectroscopy (Ch. 10 Bard & Notes)

Important Dates (subject to change)

Exam 1: In Class Exam (**Thursday October 10, 2024**)

Exam 2: In Class Exam (**Thursday December 5, 2024**)

Class Presentations: **November 21, 2024**